“Intelligent Student Performance Prediction and Academic Recommendation System”

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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2026

DECLARATION

I hereby declare that the project report entitled “Intelligent Student Performance Prediction and Academic Recommendation System” is an original work done in the Department of Computer Science and Systems Engineering, GITAM School of CSE, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the project report entitled “Intelligent Student Performance Prediction and Academic Recommendation System” is a bonafide record of work carried out by Sloka Pakala (2023000669), Sanzil Lazarus (2023000219), Tanvi Buddha (2023006493), Vishal Kemburu (2023001836) students submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering.

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1. ABSTRACT

Academic performance in higher education depends on many factors besides examination marks, such as attendance, engagement with learning resources, assessment behaviour, motivation, interests and participation in extracurricular activities. Traditional evaluation relies mainly on marks and therefore tends to identify students who need support only after difficulties have become visible. This project proposes an **Intelligent Student Performance Prediction and Academic Recommendation System** that applies machine learning, learning analytics and explainable artificial intelligence to predict student performance and to provide personalized academic guidance.

The system builds a student profile from academic, assessment, attendance and learning-engagement data, together with optional and voluntary survey information on emotional well-being and interests, collected only with approval and with privacy safeguards. Supervised models (Logistic Regression, Decision Tree, Random Forest, Support Vector Machine and regression models) are compared; SHAP is used to explain the factors behind each prediction; and an evidence-linked recommendation engine converts the findings into study and skill-development guidance, with generative AI used only to phrase the advice. Results are presented through student and faculty dashboards, and faculty can review every risk flag before any intervention. Two public datasets, the UCI Student Performance dataset and the Open University Learning Analytics Dataset (OULAD), are used for development and evaluation. The expected outcome is a reproducible prototype and an evidence-based evaluation; no claim of improved student outcomes is made before testing.

Keywords: student performance prediction, learning analytics, educational data mining, explainable AI, SHAP, academic recommendation, dashboard.

2. INTRODUCTION

2.1 BACKGROUND

Universities collect large amounts of data about their students: marks, attendance, assessment submissions, activity on learning platforms and, increasingly, information about interests and participation in co-curricular work. Educational data mining and learning analytics aim to turn such data into insight that helps teachers and students act early. Earlier studies showed that data mining techniques can predict the performance of secondary school students from school, demographic and social attributes [1], using the public UCI Student Performance dataset [2]. Large-scale learning-

analytics data, such as the Open University Learning Analytics Dataset (OULAD) [3], [4], have made it possible to study how interaction with an online learning environment relates to assessment results.

At the same time, a prediction on its own is of limited use to a faculty member or a student. Users need to know *why* a student has been flagged and *what* can be done about it. Explainable AI methods such as SHAP [5] make it possible to attribute a prediction to individual input features, which is the basis for the explanation and recommendation modules of this project.

2.2 NEED FOR THE SYSTEM

Marks alone are a late and incomplete indicator of a student's progress. Factors such as attendance, learning engagement, motivation, interests and extracurricular participation may also be relevant to understanding how a student is progressing, yet they are rarely brought together in one place. Faculty members therefore lack a centralized view of each student, and students receive generic advice rather than guidance linked to their own situation. A system that combines these indicators, predicts performance early, explains the result and suggests suitable actions can help faculty decide where support may be needed.

2.3 JUSTIFICATION OF THE TITLE

The title **"Intelligent Student Performance Prediction and Academic Recommendation System"** describes the main parts of the work:

- **Intelligent** – supervised machine learning models learn the relationship between student indicators and outcomes instead of relying on fixed rules.
- **Student Performance Prediction** – the system estimates a numerical academic score or a risk category from academic, engagement and profile data.
- **Academic Recommendation** – identified learning needs are linked to personalized study and skill-development guidance.
- **System** – prediction, explanation, recommendation and dashboards are integrated into one workflow that can be used by students and authorized faculty.

2.4 SCOPE OF THE PROJECT

The project covers the design of the system, the selection and preparation of two public datasets, the comparison of baseline prediction models, the explanation of predictions, rule-based recommendations phrased with the help of generative AI, and a prototype dashboard. Predictions

are used only as decision-support information for faculty and are not final judgments about any student.

3. LITERATURE REVIEW

3.1 PREDICTING STUDENT PERFORMANCE WITH DATA MINING

Cortez and Silva [1] applied data mining to predict the final grade of secondary school students in two Portuguese schools, using data for Mathematics and Portuguese language courses [2]. They compared several model families, including decision trees, random forests, neural networks and support vector machines, and showed that earlier period grades are the strongest predictors of the final grade. They also reported that school-related and social attributes, such as absences and previous failures, add useful information when earlier grades are not available. The study established a compact and widely used benchmark for grade prediction, and it is used in this project as the baseline prediction dataset.

3.2 LEARNING ANALYTICS IN ONLINE LEARNING ENVIRONMENTS

Kuzilek, Hlosta and Zdrahal [3] released the Open University Learning Analytics Dataset, which links anonymised student demographics, assessment results and click-stream interaction logs from a virtual learning environment across several courses and presentations [4]. Because of its size and the combination of assessment and engagement data, OULAD supports research on early identification of students at risk and on the relation between activity on a learning platform and final results. In this project it supplies the assessment and learning-engagement component that the smaller UCI dataset lacks.

3.3 EXPLAINABLE AND INTERPRETABLE MODELS

Predictive accuracy alone is not sufficient in an educational setting, because a teacher must be able to justify an intervention. Interpretable models such as Decision Trees and Logistic Regression expose their decision logic directly, while ensemble methods such as Random Forest [6] and margin-based classifiers such as the Support Vector Machine [7] can capture more complex relationships between features and outcomes. SHAP [5] provides a unified way to attribute the output of any such model to its input features, so that the most influential factors can be shown next to each prediction.

3.4 SUMMARY AND RESEARCH GAP

Table 3.1 summarizes the reviewed work and the gap that this project addresses.

Table 3.1 : Summary of reviewed work and gap addressed

Ref.	Focus / data	Contribution	Gap addressed in this project
[1], [2]	Secondary school grades; UCI Student Performance dataset	Compared data mining models for grade prediction	Adds explanations, recommendations and a dashboard on top of prediction
[3], [4]	OULAD: demographics, assessments, VLE click-stream	Public learning-analytics dataset for engagement and outcome studies	Combines engagement data with academic data in one student profile
[5]	Model-agnostic explanation (SHAP)	Feature-level attribution of predictions	Used to link predictions to evidence-based guidance

The literature shows that grade and risk prediction is well studied, but prediction models are usually reported separately from explanation and from advice to the student. Interests, extracurricular participation and emotional well-being are also rarely considered. This project therefore focuses on an integrated, explainable and privacy-aware pipeline that goes from prediction to personalized recommendation.

4. PROBLEM IDENTIFICATION & OBJECTIVES

4.1 PROBLEM IDENTIFICATION

Traditional academic evaluation mainly focuses on examination marks and may not identify students who need support early. Factors such as attendance, learning engagement, motivation, interests and extracurricular participation may also be relevant to understanding student progress, but they are generally not combined with marks in a single, explainable view. As a result, faculty see the problem late, and students do not receive guidance that is linked to their own strengths and needs.

4.2 AIM OF THE PROJECT

To develop an intelligent system that predicts student performance and provides personalized academic guidance using machine learning, learning analytics and explainable AI.

4.3 OBJECTIVES

- **Objective 1 – Comprehensive student performance analysis:** to collect and analyse academic, assessment, attendance and learning-engagement data, and to incorporate relevant student-reported emotional well-being, interests and extracurricular information where

available, in order to understand the factors associated with student performance and engagement.

- **Objective 2 – Explainable prediction and personalized recommendations:** to develop an explainable prediction and recommendation system that identifies students who may benefit from support, presents the relevant factors through a centralized dashboard, and provides personalized academic and skill-development guidance.

4.4 FEEDBACK FROM REVIEW 1 AND ACTIONS TAKEN

The feedback received from the guide and reviewers in Review 1 [9] was used to widen the scope of the project. Table 4.1 lists the feedback and the corresponding actions.

Table 4.1 : Review 1 feedback and actions taken / planned

S.No.	Feedback received in Review 1	Action taken / planned
1	Consider emotional well-being, motivation and student interest alongside academic marks.	Explore additional voluntary student surveys for emotional well-being and interests, subject to approval.
2	Include extracurricular activities, skills and student participation.	Expand the student profile to include academic, engagement and personal-development indicators.
3	Provide a centralized dashboard where faculty can view student information.	Design a dashboard to help faculty understand student progress and potential support needs.
4	Identify factors affecting academic performance, engagement and overall student efficiency.	Use two public datasets for academic performance and learning-engagement analysis; develop explainable ML models and personalized recommendations.

4.5 REQUIREMENT ANALYSIS

The requirements derived from the objectives and from the Review 1 feedback are given in Table 4.2.

Table 4.2 : Functional and non-functional requirements

Functional requirements	Non-functional requirements
Collect student academic data.	High prediction accuracy.
Track learning and engagement.	Secure data handling.
Analyse interests and extracurricular activities.	Protection of student privacy.
Consider emotional well-being indicators.	User-friendly interface.
Predict student performance.	Fast system response.
Provide recommendations through dashboards.	Fair and explainable predictions.

5. EXISTING SYSTEM

In most institutions and in much of the published work, student progress is monitored through the following approaches:

- **Mark-based evaluation:** internal and end-semester marks are reviewed after assessments are completed.
- **Attendance and learning-platform reports:** attendance registers and learning-platform logs record what happened, but are usually kept in separate systems and are rarely combined with marks.
- **Stand-alone grade-prediction models:** research prototypes such as the models of [1] predict grades from school or demographic data, but normally stop at the prediction.
- **General, non-personalized advice:** counselling and study guidance are given to a class or to individual students on the basis of a teacher's own observation.

Table 5.1 summarizes these approaches and their drawbacks.

Table 5.1 : Existing approaches and their drawbacks

Existing approach	Main drawback
Mark-based evaluation	Reactive: students needing support may be identified only after poor results appear.
Separate attendance and platform reports	Indicators are scattered; no combined student profile or centralized faculty view.
Stand-alone prediction models	Often a "black box": no explanation of influencing factors and no suggested action.
Generic advice	Not linked to the student's own data, interests or skills.
Marks-centred data	Motivation, interests, extracurricular participation and well-being are generally not considered.

In summary, the existing arrangements describe what has already happened, rather than combine the relevant indicators, explain the factors behind a prediction and connect them to a practical next step.

6. PROPOSED SYSTEM

6.1 OVERVIEW

The proposed system connects performance prediction, explanation and personalized academic recommendation in one workflow. It accepts academic records, learning-engagement data and optional student self-assessment information, validates and preprocesses them, estimates an academic score or risk level, identifies the factors that influence the result, and generates guidance that students and authorized faculty can view on dashboards.

6.2 MODULES OF THE PROPOSED SYSTEM

- **Data acquisition and validation:** loads academic, engagement and optional survey data and checks them for missing values, class imbalance and leakage.
- **Preprocessing and student profile:** cleans the data and engineers features that form the academic, engagement and personal-development profile.
- **Prediction module:** estimates a numerical academic score (regression) or a risk category (classification) using the compared models.
- **Explanation module:** uses SHAP and feature importance to show which factors contributed to each prediction.
- **Recommendation engine:** maps identified learning needs to personalized study and skill-development actions using evidence-linked rules; generative AI may phrase the advice but must not invent facts.
- **Dashboards:** a student dashboard for individual progress and a faculty dashboard for student profiles and cohort-level patterns.

6.3 DATASETS

Two public datasets are used for academic performance and learning-engagement analysis (Table 6.1). They are public or synthetic prototype data and are not presented as genuine observations of GITAM students.

Table 6.1 : Datasets used in the project

Dataset	Content	Use in the project
UCI Student Performance [2]	Secondary school students of two Portuguese courses (Mathematics and Portuguese language): grades, absences, demographic and social attributes	Compact grade-prediction baseline; regression and classification experiments
Open University Learning Analytics Dataset (OULAD) [3], [4]	Anonymised student demographics, assessment results and virtual learning environment click-stream data	Assessment and learning-engagement analysis; risk classification

6.4 PROPOSED METHODOLOGY

The methodology follows three layers, which are shown in Figure 7.1 and described below.

- **Input layer:** grades, attendance/absences, assessment submissions and available learning-platform activity, with optional voluntary survey data.

- **Decision layer:** predicted score or risk, the confidence of the prediction and the main contributing features.
- **Support layer:** evidence-linked actions; generative AI may phrase the advice but should not invent facts. Faculty can review risk flags and recommendations before any intervention.

6.5 EVALUATION PLAN

Models will be compared with metrics that suit the prediction target (Table 6.2). The tests will check missing values, class imbalance, data leakage, subgroup errors and whether the features exist at the selected prediction time.

Table 6.2 : Prediction tasks, candidate models and metrics

Task	Candidate models	Evaluation metrics
Numerical academic score	Regression models	MAE, RMSE
Risk / outcome classification	Logistic Regression (baseline), Decision Tree, Random Forest, Support Vector Machine	Precision, recall, F1, ROC-AUC
Explanation of predictions	SHAP, feature importance	Consistency and usefulness of influencing factors for faculty

6.6 IMPROVEMENT OVER THE EXISTING SYSTEM

Table 6.3 : Comparison of the existing and the proposed system

Aspect	Existing system	Proposed system
Data used	Marks, attendance and platform logs kept separately	Academic, engagement and optional personal-development indicators in one profile
Timing	After assessments are completed	Predicts score or risk so that support can be considered early
Explanation	Little or none	SHAP-based factors shown with each prediction
Guidance	Generic advice	Personalized, evidence-linked recommendations
Faculty view	Scattered reports	Centralized dashboard with student and cohort views
Oversight	Not defined	Faculty review before any intervention; privacy safeguards

7. SYSTEM ARCHITECTURE

7.1 BLOCK DIAGRAM

Figure 7.1 shows the block diagram of the system. Student and course data enter the input layer. The decision layer validates and preprocesses the data, builds the student profile, predicts the score or risk and explains the result. The support layer turns the findings into recommendations and presents them on the student and faculty dashboards. Faculty oversight and privacy safeguards apply across all layers. The workflow, data flow, sequence, activity and use case diagrams in the following sections describe the system from different points of view.

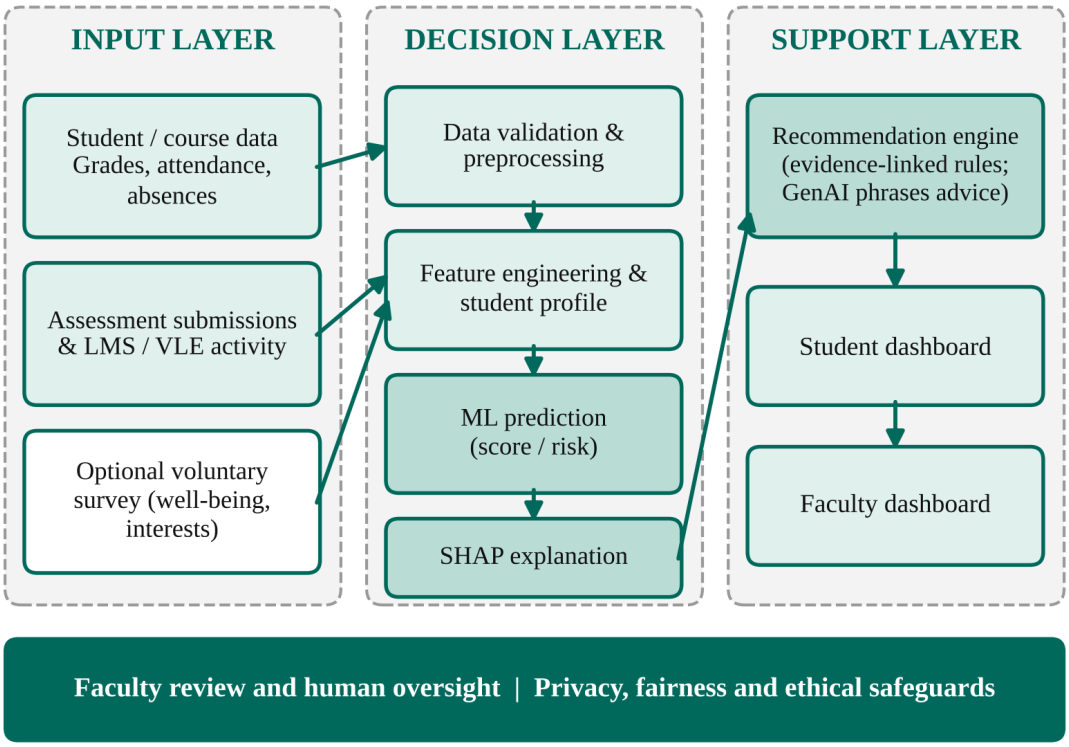
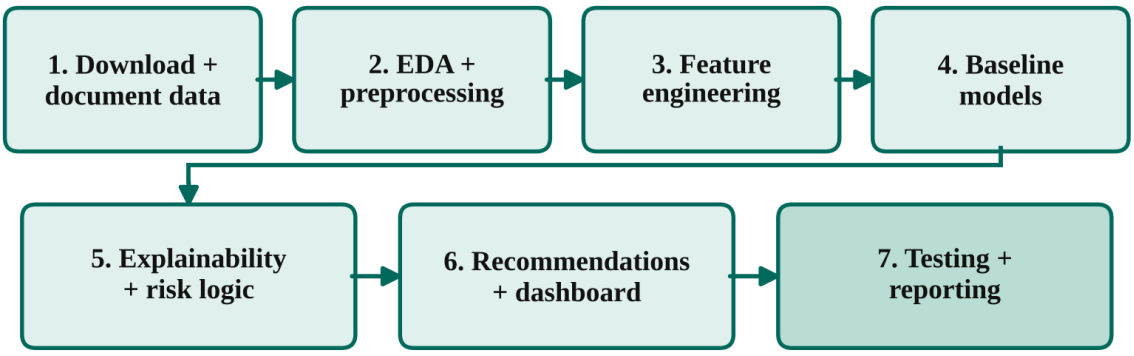


Figure 7.1 : System architecture of the proposed system

7.2 PROJECT WORKFLOW

The implementation proceeds in seven phases, from dataset set-up to final evaluation (Figure 7.2).



Validation checkpoints apply at every phase. Faculty can review risk flags and recommendations before any intervention.

Figure 7.2 : Proposed project workflow

- **Validation checkpoints:** check missingness, class imbalance, leakage and subgroup errors, and whether the features exist at the selected prediction time.
- **Expected deliverables:** documented datasets, reproducible notebooks, an evaluation table, explanation examples, recommendation rules and a dashboard prototype.

7.3 DATA FLOW DIAGRAM (DFD)

The system receives academic records, learning-engagement data and optional student self-assessment information. The data are validated and processed before being passed to the prediction model. Explainability techniques identify influential factors, and the recommendation engine generates personalized academic guidance. Students can view their own progress, while authorized faculty can review relevant student profiles and cohort-level patterns (Figure 7.3).

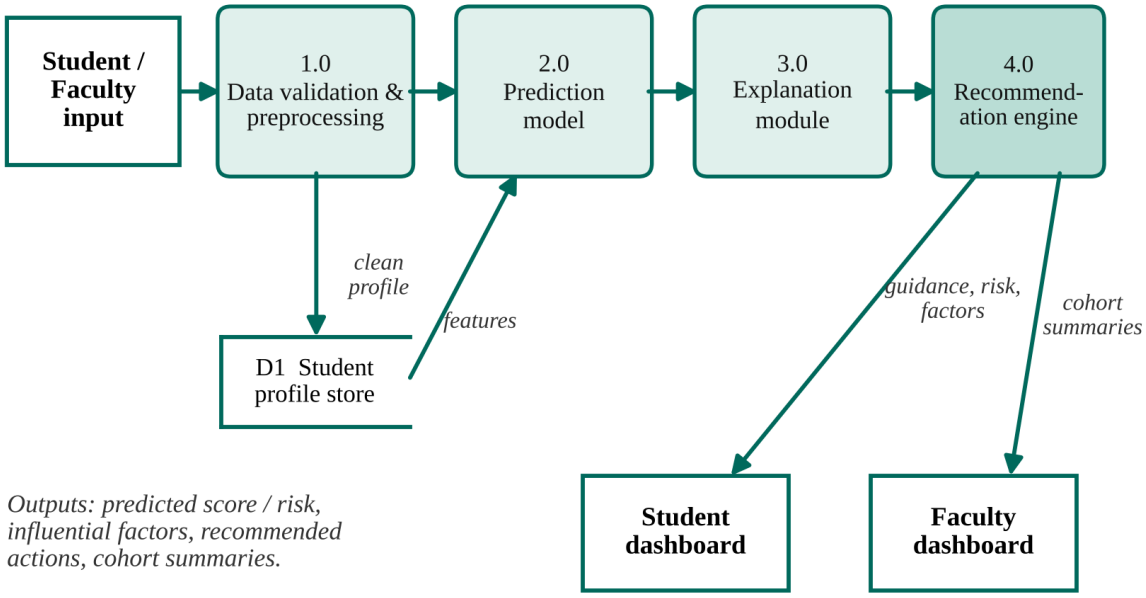


Figure 7.3 : Data flow diagram of the proposed system

The main components of the data flow are described in Table 7.1.

Table 7.1 : Main components of the data flow

Component	Description
Input	Academic, engagement and optional survey data
Processing	Validation, preprocessing and student-profile creation
Prediction	Academic score or risk estimation
Analysis	Explainability and student-development indicators
Recommendation	Personalized study and skill-development guidance
Output	Student and faculty dashboards

7.4 SEQUENCE DIAGRAM

Figure 7.4 shows the interaction between the user (student or faculty), the Streamlit frontend dashboard, the Python processing and machine learning engine, and the database / dataset repository. After the user logs in and opens a dashboard view, the frontend requests the student records and metrics from the backend. The backend queries the academic and engagement history from the repository, preprocesses the data and engineers features, runs the prediction model, computes the SHAP explanations, and builds the risk flags and recommendations (steps 3 to 9).

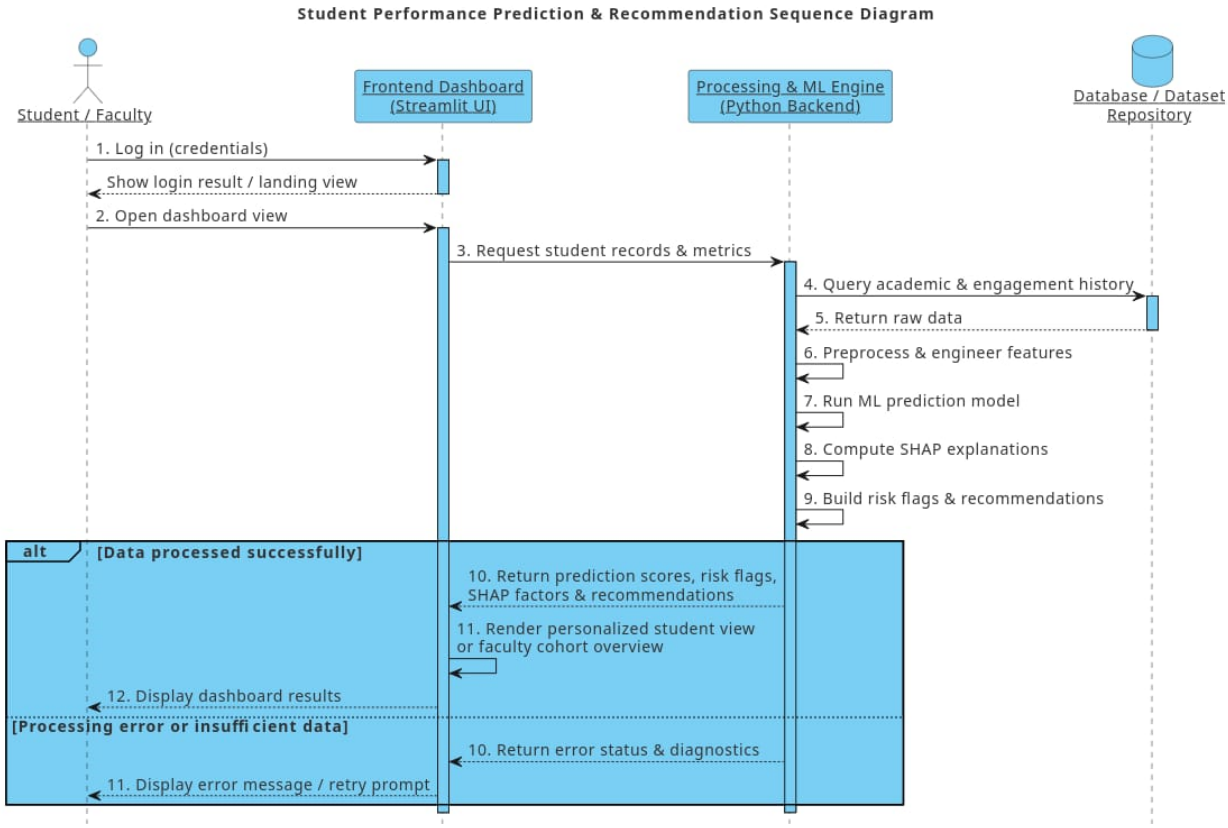


Figure 7.4 : Sequence diagram of student performance prediction and recommendation

If the data are processed successfully, the prediction scores, risk flags, SHAP factors and recommendations are returned to the frontend, which renders a personalized student view or a faculty cohort overview. If processing fails or the data are insufficient, an error status with diagnostics is returned and the user sees an error message with a retry prompt.

7.5 ACTIVITY DIAGRAM

The activity diagram in Figure 7.5 follows a record through the pipeline, with one swim-lane for each stage: input layer, data validation and preprocessing, prediction model, explanation module, recommendation engine, and decision layer and dashboards.

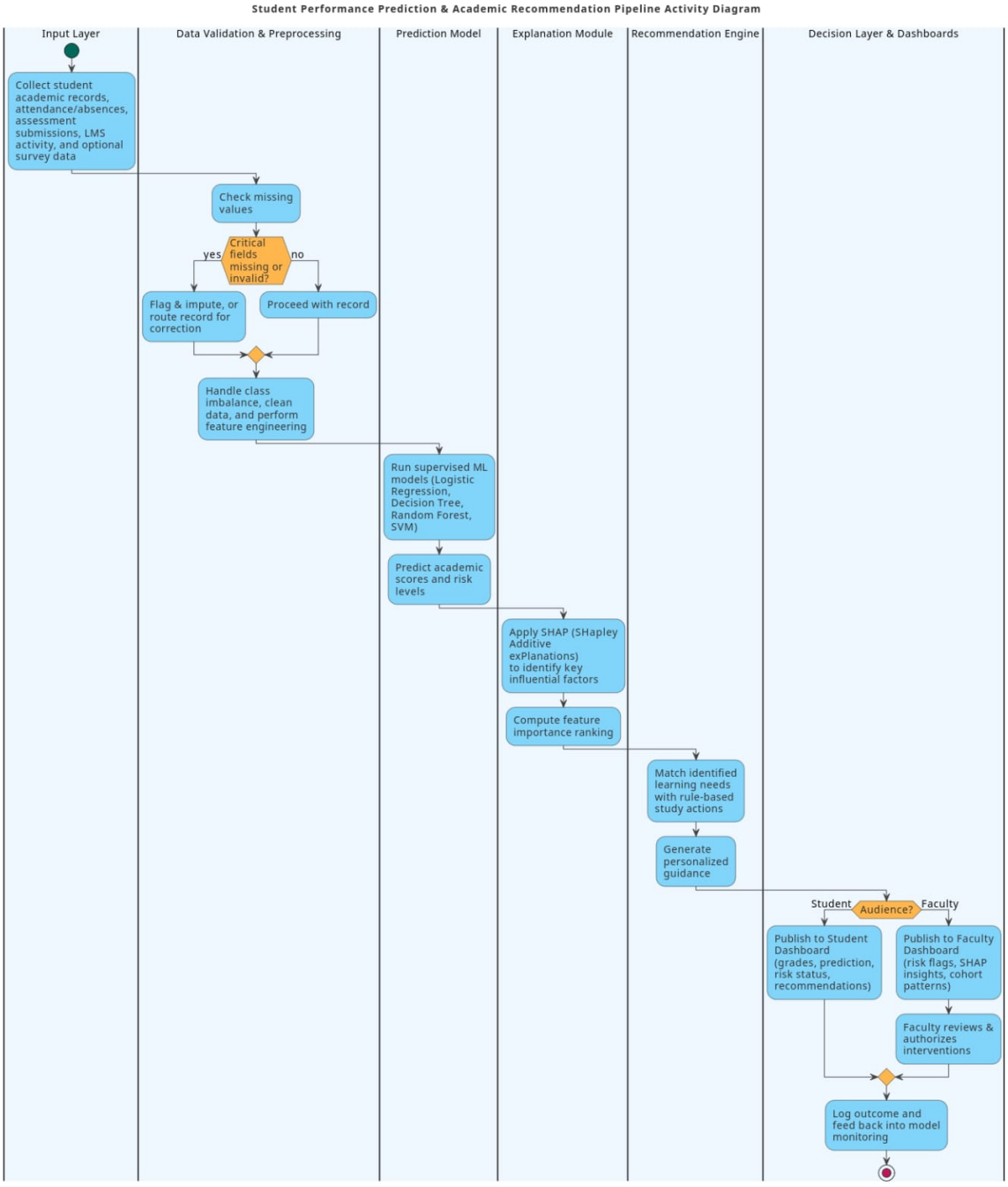


Figure 7.5 : Activity diagram of the prediction and recommendation pipeline

- **Input and validation:** academic records, attendance/absences, assessment submissions, LMS activity and optional survey data are collected and checked for missing values. A record with critical fields missing or invalid is flagged and imputed, or routed for correction; otherwise it proceeds. Class imbalance is then handled, the data are cleaned and features are engineered.

- **Prediction and explanation:** the supervised models (Logistic Regression, Decision Tree, Random Forest, SVM) predict academic scores and risk levels; SHAP then identifies the key influential factors and a feature-importance ranking is computed.
- **Recommendation and dashboards:** identified learning needs are matched with rule-based study actions and personalized guidance is generated. Output is published to the student dashboard (grades, prediction, risk status, recommendations) or to the faculty dashboard (risk flags, SHAP insights, cohort patterns), where faculty review and authorize interventions. The outcome is logged and fed back into model monitoring.

7.6 USE CASE DIAGRAM

The use case diagram in Figure 7.6 shows the actors of the system and the functions available to them. The student and the faculty member are the primary actors; the administrator is a secondary actor who looks after the data and models.

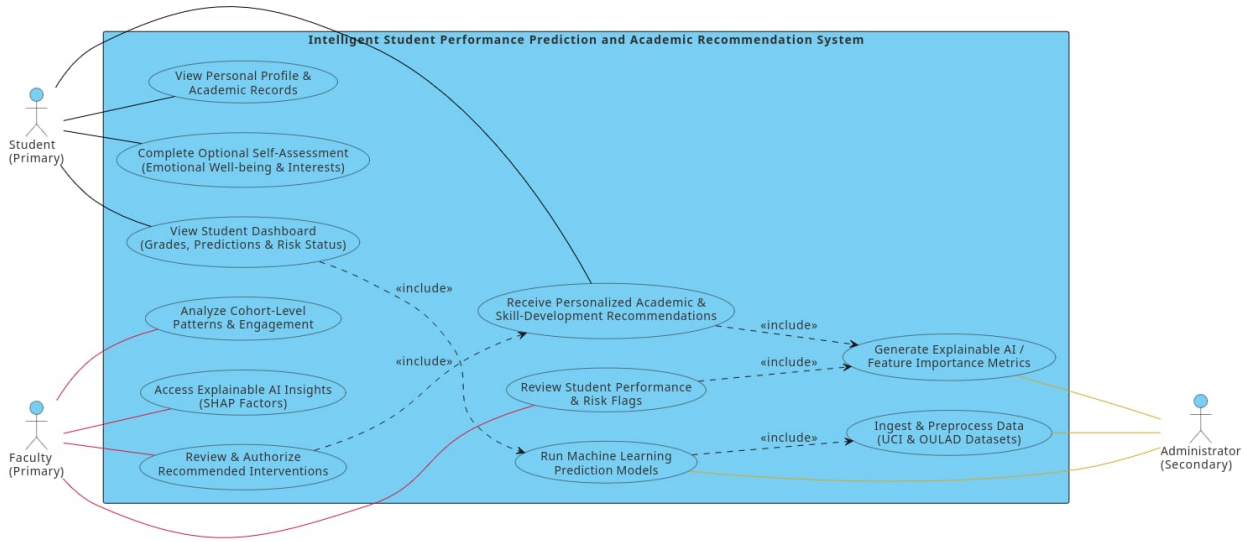


Figure 7.6 : Use case diagram of the proposed system

- **Student:** views the personal profile and academic records, completes the optional self-assessment (emotional well-being and interests), views the student dashboard (grades, predictions and risk status) and receives personalized academic and skill-development recommendations.
- **Faculty:** analyzes cohort-level patterns and engagement, accesses explainable AI insights (SHAP factors), reviews student performance and risk flags, and reviews and authorizes the recommended interventions.

- **Administrator:** ingests and preprocesses the UCI and OULAD datasets, runs the machine learning prediction models and generates the explainable AI / feature-importance metrics on which the other use cases depend ("include" relationships).

8. TOOLS/TECHNOLOGIES USED

8.1 TOOLS

Table 8.1 lists the software tools and their significance in the project.

Table 8.1 : Tools used and their significance

Tool	Significance in the project
Python	Programming and model development
Pandas and NumPy	Data loading, cleaning, processing and feature engineering
Jupyter Notebook / Google Colab	Experimentation and reproducible notebooks
Scikit-learn [8]	Machine learning models, pipelines and evaluation metrics
SHAP [5]	Explainability of model predictions
Streamlit	Development of the student and faculty dashboards
Git and GitHub	Version control and team collaboration

8.2 TECHNOLOGIES

The project is based on the following technologies:

- **Educational Data Mining and Learning Analytics** – to analyse academic, assessment and engagement data.
- **Supervised Machine Learning** – to learn the relationship between student indicators and outcomes.
- **Explainable Artificial Intelligence** – to show the factors behind each prediction.
- **Generative AI for personalized guidance** – to phrase advice that is based on identified evidence.
- **Data visualization and dashboard development** – to present results to students and faculty.

8.3 ALGORITHMS AND ANALYTICAL METHODS

Table 8.2 : Algorithms and analytical methods

Algorithm / method	Role in the project
Logistic Regression	Baseline risk classification
Decision Tree	Interpretable prediction rules
Random Forest [6]	Captures complex relationships between features and outcomes
Support Vector Machine [7]	Candidate classification model
Regression models	Predict numerical academic scores
SHAP [5]	Explains the important factors influencing predictions
Rule-based recommendations	Connect identified learning needs to suitable academic support

9. CONCLUSION

9.1 CONCLUSION

This report presented the proposed Intelligent Student Performance Prediction and Academic Recommendation System. The system connects performance prediction, explanations and personalized academic recommendations. The UCI Student Performance dataset provides a compact grade-prediction baseline, and OULAD adds assessment and learning-platform engagement data. Models will be compared with metrics appropriate to the target: MAE and RMSE for scores, and precision, recall, F1 and ROC-AUC for risk classification. Students and faculty will receive understandable outputs, with privacy safeguards and human oversight. The expected result is a reproducible prototype and an evidence-based evaluation, not a claim of improved outcomes before testing.

9.2 WORK PLAN AND TIMELINE

The proposed work plan for October–November is given in Table 9.1; task status will be updated to reflect actual progress.

Table 9.1 : Work plan and timeline

Task	Task owner	Status	Target
Dataset download and data dictionary	Sanzil, Tanvi	Planned / verify	Oct
EDA, preprocessing and features	Sanzil, Tanvi	Planned	Oct
Baseline models and evaluation	Vishal, Sloka	Planned	Oct–Nov
Explainability and risk interpretation	Tanvi	Planned	Nov
Recommendations and GenAI integration	Sloka, Vishal	Planned	Nov
Dashboard, testing and report	All	Planned	Nov

9.3 DISCLOSURE STATEMENTS

- We will maintain academic honesty by avoiding plagiarism, cheating, unauthorized collaboration and the falsification of data or results.
- We will properly acknowledge and cite all research papers, datasets, software tools and AI tools used in this project.
- We will clearly distinguish original public datasets from synthetic prototype data and will not present generated records as genuine student observations.
- We will respect student privacy and confidentiality, obtain appropriate permission for any additional data collection, and ensure that emotional well-being information is handled responsibly.
- We will use machine learning predictions as decision-support information rather than definitive judgments about students. We will verify all generated content, code, experimental results and conclusions before including them in the final project report or presentation.

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